

Vantage: Project Proposal

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(AAI4001): Supervised Machine Learning

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Problem Definition

Real-World, Industry-Relevant Problem

Tourism is one of Jamaica's most important economic sectors, and visitor arrivals directly influence national revenue, employment, and business planning. However, tourism agencies and stakeholders currently rely heavily on historical trends without predictive tools that can forecast future monthly visitor arrivals by market or country of origin. This limits the ability to plan resources, marketing campaigns, and infrastructure effectively.

Importance

This problem is important because accurate visitor-arrival predictions help:

- Tourism Agencies (e.g., JTB) – target promotional campaigns toward high-growth markets and prepare for seasonal demand changes.
- Businesses (hotels, transport, attractions) – optimize staffing, inventory, and operational planning.
- Government & Planners – anticipate economic impact and allocate resources more efficiently.

Predictive insights support stronger decision-making across the tourism sector and help maximize Jamaica's economic performance.

Description of Dataset

Source

The dataset was obtained from Jamaica's Open Data Portal (data.gov.jm). It was compiled and published by tourism-related government agencies, including the Jamaica Tourist Board (JTB), which tracks visitor arrivals by country and port of entry.

Dataset Link: Purpose of Visit by Port and Country – JTB Open Data Request

<https://data.gov.jm/sites/default/files/Purpose-of-Visit-by-Port-and-Country-JTB-Open-data-request.csv>

Data Coverage

- Countries - It shows where visitors to Jamaica come from.
- Time: The data is collected over multiple years, usually from 2014 onwards.
- Visitors included: People who stay in Jamaica for at least one night (stopover visitors), including Jamaicans living abroad who come back to visit.
- Visitors excluded: Cruise passengers who don't stay overnight aren't counted.

Features (i.e, independent variables)

- Country visitors come from.
- Port of entry (like which airport or sea port they arrive at).
- Year of arrival

Labels (i.e.target variables)

- Number of visitors from each country.
- Purpose of their visit (like leisure, business, or other reasons)

Dataset Size

This dataset has 19 thousand records. There are lots of rows because it tracks multiple countries over multiple years, and also breaks down visitors by purpose.

Limitations

Some limitations are:

- Doesn't include some visitors, like cruise passengers who don't stay overnight.
- The data depends on what visitors report on their arrival forms, so there could be mistaken or missing info.
- Non-residents Jamaicans are included, which might affect counts if we're looking for foreign visitors
- Some years might have incomplete data.
- The “purpose of visit” categories are broad and might not capture all details

Proposed Approach**Algorithms & Rationale**

In order to address the task of predicting monthly tourism arrivals in Jamaica, the project will have to train and compare various machine learning algorithms that are suitable to the dataset. The proposed algorithms balance interpretability, validity, and overall performance to align with constraints of the dataset.

1. Linear Regression Algorithm***Rationale***

Linear Regression will act as the starting/baseline model because of its simplicity, interpretability, and effectiveness for capturing linear trends and seasonal patterns in tourist arrivals. It also provides clear insight into which variables would most strongly influence travel counts.

Advantages

High interpretability, effective for identifying linear trends, seasonality, and small or moderate time series, small risk of overfitting.

Disadvantages

Sensitive to outliers, limited capacity for collecting complex/ non-linear interactions.

2. Random Forest (Classification Algorithm)***Rationale***

The Random Forest method could be used as the primary predictive model for the dataset because it is appropriate for recording non-linear relationships that might be present in the tourism arrival dataset, as they sometimes follow non-linear seasonal patterns

influenced by various factors. The ensemble structure, which makes use of multiple decision trees, allows for the formation of complex patterns without introducing linear assumptions. Random Forests are reliable with noisy real-world datasets as well as provide techniques useful for feature analysis. therefore supporting interpretability requirements.

Advantages

Can model non-linear patterns & feature interactions, resistant to overfitting, supports interpretability through feature importance mechanisms.

Disadvantages

Less interpretable than linear models & requires more computational resources.

Evaluation Metrics

In order to properly assess and compare the performance of the selected algorithms to ensure proper evaluation for accuracy, risk sensitivity & real-world interpretability, these regression metrics can be used:

- Mean Absolute Error: To measure the average difference between the predicted and actual monthly visitor arrival, providing a more interpretable measure of accuracy which reflects the typical size of the prediction errors.
- Root Mean Squared Error: Assesses larger errors heavily making it suitable for evaluating the model's performance on months with unusually high or low tourism activity. This would be helpful in identifying how well the model would handle peak seasons and sudden changes.
- Coefficient of Determination (R² Score): Indicates the proportion of variation in visitor arrivals explained by the model. A higher R² source would represent a better overall fit and a stronger predictive capability.

Expected Outcomes

At the end of this project we expect to produce:

- A fully analysed, processed and cleaned dataset with visualisations highlighting key visitor trends, market behaviours and seasonal patterns.
- A predictive tool capable of estimating future monthly visitor arrivals based on different markets using historical tourism patterns.
- Trained machine learning models (Random Forest & Linear Regression) comparing accuracy, performance & interpretability.
- Actionable Insights: Seasonal demand peaks, high-growth markets, and feature important rankings that identify the major variables influencing visitor arrivals.